

Oriented chromatic number of planar graphs

OPG-494

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

An oriented colouring of an oriented graph is assignment c of colours to the vertices such that no two arcs receive ordered pairs of colours (c_1, c_2) and (c_2, c_1) . It is equivalent to a homomorphism of the digraph onto some tournament of order k .

Problem What is the maximal possible oriented chromatic number of an oriented planar graph?